CAREER-VISTA
JOB ROLE RECOMMENDATION SYSTEM

Mini Project Report

Submitted for the Partial fulfilment of the Requirements

for the Award of the Degree of

MASTER OF COMPUTER APPLICATIONS

By

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October-2025

Department of Computer Applications

BONAFIDE CERTIFICATE

This is to certify that the Mini Project Report entitled "……………………………… ……………………………" has been submitted by Mr./Ms ………………………………… Reg. No. ……………………… …for the partial fulfilment of the requirements for the award of the degree of Master of Computer Applications (MCA) of APJ Abdul Kalam Technological University, Kerala during the year 2025

Place: Varikoli

Date:

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Submitted for the Final Evaluation held on ………………………

Name and Signature of the Examiner

DECLARATION

I, undersigned hereby declare that the Mini Project Report ***"Carrer-Vista Job Role Recommendation System"****, submitted for the partial fulfilment of the requirements for the award of degree of Master of Computer Applications of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under the supervision of Sivadas T Nair . This submission represents my ideas in my own words and where ideas or words of others also have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.*

Gokul Raj C

MUT24MCA-2029

Place:

Date:

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ABSTRACT

The Career-Vista Job Role Recommendation System is a web-based intelligent application developed to recommend the most suitable job roles for users based on their technical and personal skillsets. The system integrates knowledge from domains such as Database Fundamentals, Computer Architecture, Distributed Computing Systems, Cyber Security, Networking, Software Engineering, and Data Science to provide accurate and data-driven role suggestions.

The frontend is designed using HTML and CSS, offering an intuitive and responsive user interface. The backend is implemented using Python Flask, ensuring smooth communication between client and server, while MongoDB is used for storing user data and managing secure sign-in and sign-up functionalities.

By analyzing user attributes such as programming skills, analytical thinking, and technical knowledge, the system predicts suitable roles including AI/ML Specialist, Database Administrator, Application Support Engineer, Hardware Engineer, Networking Engineer, Software Developer, API Specialist, Cyber Security Specialist, Project Manager, Information Security Specialist, Technical Writer, Software Tester, Business Analyst, Customer Service Executive, Helpdesk Engineer, and Graphics Designer.

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LIST OF ABBREVATIONS

ABBREVATIONS	DESCRIPTION
AI	Artificial Intelligence
ML	Machine Learning
API	Application Programming Interface
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
CSV	Comma-Separated Values
GPU	Graphics Processing Unit
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
RFC	Random Forest Classifier
VSC	Visual Studio Code
URL	Uniform Resource Locator

CHAPTER 1

INTRODUCTION

In today's rapidly evolving technological landscape, selecting an appropriate job role that aligns with an individual's technical skills, academic background, and interests has become increasingly challenging. With the expansion of domains such as Artificial Intelligence, Cyber Security, Data Science, and Software Engineering, students and professionals are often uncertain about which job role best suits their skillsets and career goals. Traditional career guidance methods rely heavily on manual assessments, generalized aptitude tests, or counselor recommendations, which may lack accuracy and personalization.

The Career-Vista Job Role Recommendation System aims to bridge this gap by leveraging data-driven insights to suggest the most suitable job roles for users based on their academic knowledge, technical competencies, and soft skills. Unlike traditional career recommendation platforms, this system uses a structured evaluation of user inputs across multiple computing disciplines—such as Database Fundamentals, Distributed Systems, Cyber Security, Networking, Software Development, and Project Management—to deliver precise job role predictions. The system provides a secure and user-friendly environment where users can sign up and log in using a Flask and MongoDB-based backend, ensuring efficient data handling and authentication.

By analyzing a user's proficiency across multiple domains, the system recommends potential roles including AI/ML Specialist, Database Administrator, Application Support Engineer, Hardware Engineer, Networking Engineer, Software Developer, API Specialist, Cyber Security Specialist, Project Manager, Information Security Specialist, Technical Writer, Software Tester, Business Analyst, Customer Service Executive, Helpdesk Engineer, and Graphics Designer. The ultimate goal of Career-Vista is to empower individuals with personalized insights, helping them make informed career decisions and align their strengths with the evolving demands of the technology industry.

1.1 BACKGROUND & MOTIVATION

With rapid advancements in computing and digital technologies, industries now demand professionals with multidisciplinary skillsets and adaptability. However, students and early-career professionals often struggle to understand how their unique combination of skills academic exposure translates into specific job roles. Educational institutions typically emphasize theoretical learning in areas such as Computer Architecture, Database Systems, Networking, and Software Engineering, but provide limited guidance on how these competencies align with real-world professions.

This growing mismatch between academic knowledge and industry expectations motivates the development of an intelligent job role recommendation system. Career-Vista addresses this need by using an AI-driven approach to analyze technical and interpersonal skills, offering clear insights into job opportunities that match a user’s strengths. Furthermore, by incorporating Python Flask for backend processing and MongoDB for user management, the system ensures a seamless, secure, and scalable web application architecture.

In addition to its technical foundation, the project emphasizes communication skills, business analysis, and troubleshooting abilities—key attributes for career success in the IT industry. By integrating these elements, Career-Vista not only guides users toward suitable job roles but also promotes a better understanding of how academic subjects and personal skills combine to define professional potential.

1.2. PROBLEM STATEMENT

Despite the availability of numerous career guidance platforms, many existing systems fail to offer personalized job role recommendations based on a comprehensive evaluation of technical and interpersonal skills. Most tools provide generic suggestions without considering specific competencies across computing disciplines. As a result, students and professionals often struggle to identify career paths that truly align with their abilities and aspirations.

Moreover, traditional systems frequently lack secure and efficient user management, resulting in limited accessibility and poor user experience. There is a need for a web-based intelligent system that integrates Flask and MongoDB for authentication and data storage while applying AI-based mapping between user skills and job roles.

The Career-Vista Job Role Recommendation System aims to solve these challenges by providing a secure platform that analyzes users’ skill profiles and recommends the most suitable job roles from a predefined set of possibilities. The system’s objective is to enhance decision-making, bridge the gap between academic learning and employment readiness, and support individuals in choosing job roles that align with both their knowledge base and industry trends.

1.3 OBJECTIVES OF CAREER-VISTA

The primary objective of this project is to develop an intelligent web-based system that recommends the most suitable job roles for users based on their technical skills, academic background, and personal attributes. The system aims to provide personalized job role predictions using an AI-driven model that analyzes user inputs across various computing domains such as Database Fundamentals, Distributed Systems, Cyber Security, Networking, and Software Engineering.

One of the key objectives is to design a secure and user-friendly authentication system using Python Flask and MongoDB for managing sign-in and sign-up functionalities. This ensures that user data is stored securely while maintaining accessibility and scalability.

In addition to job role prediction, the system seeks to educate users about the skill requirements and responsibilities associated with each recommended role, helping them better understand the connection between their competencies and professional opportunities. The project also focuses on integrating a simple and responsive interface built with HTML and CSS, allowing users to interact seamlessly with the system.

Furthermore, Career-Vista aims to promote awareness of multidisciplinary skill development by highlighting the importance of both technical and soft skills such as programming ability, communication, project management, and analytical thinking. Through these capabilities, the system bridges the gap between academic learning and industry expectations, empowering users to make data-driven and well-informed career decisions.

By achieving these objectives, the Career-Vista Job Role Recommendation System provides a comprehensive and intelligent platform that assists individuals in identifying the most fitting job roles—such as AI/ML Specialist, Software Developer, Cyber Security Specialist, Database Administrator, Project Manager, and more—based on their unique skills and strengths.

1.4 ALGORITHM

The Random Forest Classifier is a powerful and versatile machine learning algorithm commonly used for classification tasks, such as predicting different job roles. It operates by building multiple decision trees during training and combining their outputs to improve accuracy and reduce overfitting. Each tree in the forest is trained on a random subset of the data and a random selection of features, which ensures that the model captures a wide variety of patterns within the dataset. When making predictions, the Random Forest takes the majority vote from all the individual trees, leading to more stable and reliable results.

Before training, the dataset is preprocessed by handling missing values, converting categorical data into numerical form, and normalizing features if necessary. This ensures that the model can interpret the data effectively. The Random Forest algorithm is highly robust to noise and performs well even with complex and nonlinear relationships in the data.

Once trained, the model's performance is evaluated using metrics such as accuracy, precision, recall, and F1-score to determine its effectiveness. In this project, the Random Forest Classifier demonstrated high prediction accuracy for job role recommendation, making it a valuable tool for supporting students and educators in making early, data-driven, and informed decisions about suitable job roles.

1.5 SCRUM METHODOLOGY

The project was developed using the Scrum framework, an Agile methodology that is particularly well-suited for projects requiring iterative development and constant feedback. Scrum provides a structured yet flexible approach to managing complex software development projects, allowing teams to deliver high-quality results through continuous collaboration and refinement.

1.5.1 Overview of the Scrum Framework

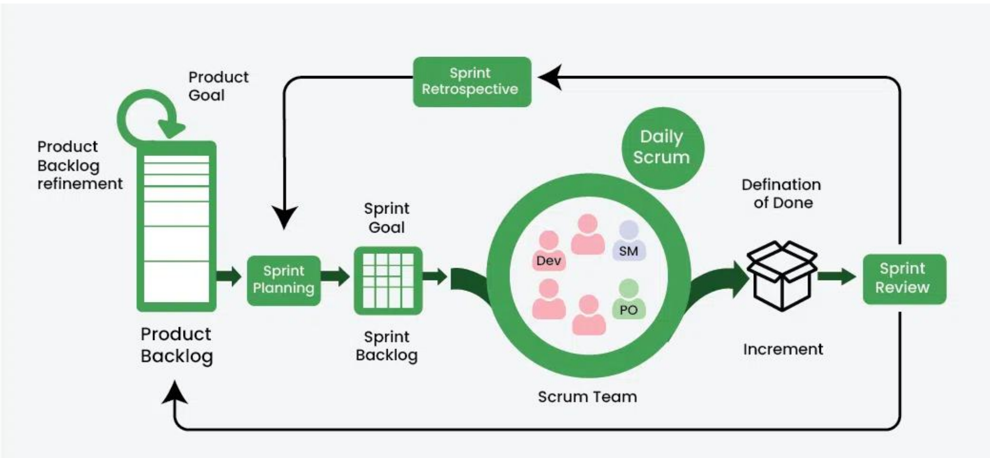

Fig 1.1 Scrum Framework

Scrum is organized around fixed-length iterations called sprints, typically lasting one to four weeks. Each sprint aims to deliver a potentially shippable product increment that stakeholders can evaluate. This approach enables teams to adapt to changes swiftly, respond to feedback, and continuously improve the product. Scrum's iterative nature also minimizes risks by ensuring that issues or changes in requirements are identified early in the development cycle.

Key roles within the Scrum framework include:

- Product Owner: Defines and prioritizes features in the product backlog based on stakeholder needs and ensures alignment with the project vision.
- Scrum Master: Facilitates Scrum ceremonies, removes obstacles, and ensures that the team follows Scrum practices.
- Development Team: Responsible for delivering the work committed to each sprint, such as coding, testing, and deployment.

1.5.2 Implementation of Scrum in the Project

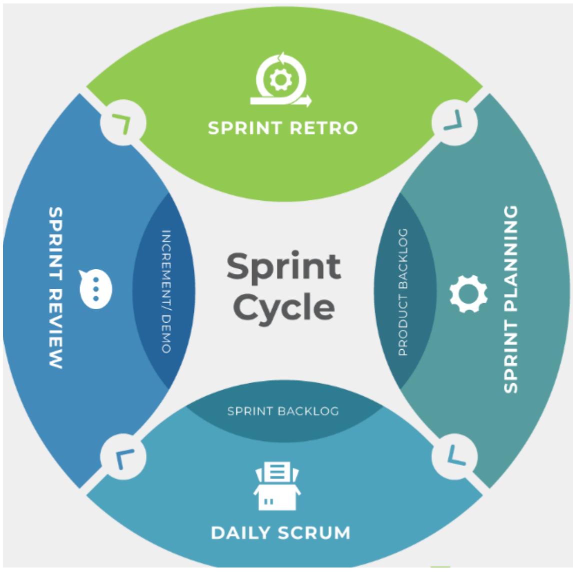

Fig 1.2 Agile Sprint Cycle

In this project, the Scrum framework was tailored to support effective progress tracking, rapid prototyping, and regular feedback incorporation. Here's how it was implemented across the development cycle:

1. Sprint Planning: At the beginning of each sprint, the team held a sprint planning session to review the project backlog, prioritize tasks, and set sprint goals. Tasks were broken down into manageable items, assigned story points based on complexity, and selected for the sprint backlog. This ensured that each sprint had a clear objective and that the workload was feasible within the sprint's timeframe.

2. Daily Stand-Up Meetings: Daily stand-ups were held every morning to discuss progress, blockers, and upcoming tasks. These brief meetings, typically lasting 15 minutes, allowed team members to stay aligned and quickly address any issues impeding progress. By maintaining open communication and transparency, the team minimized misunderstandings and fostered a collaborative environment.

3. Backlog Refinement Sessions: The product backlog was revisited regularly in backlog refinement sessions. This process involved evaluating new requirements, reprioritizing tasks, and adding details to user stories to clarify objectives. By refining the backlog consistently, the team could respond to new insights, adjust the project roadmap, and ensure that high-priority tasks were always well-defined for upcoming sprints.

4. Sprint Review: At the end of each sprint, a sprint review was conducted to demonstrate the work completed. This allowed stakeholders to view the latest product increment and provide feedback on features or adjustments needed. The review ensured that any necessary changes were identified and scheduled for the next sprint, supporting continuous improvement and alignment with stakeholder expectations.

5. Sprint Retrospective: After the sprint review, a sprint retrospective was held to reflect on the sprint process itself. This ceremony provided a platform for the team to discuss what went well, what could be improved, and how to apply lessons learned in the following sprint. By embracing a culture of continuous improvement, the team could fine-tune its processes to boost efficiency and quality in each successive sprint.

6. Continuous Feedback Loop: Feedback was collected and incorporated throughout the project, enabling rapid prototyping and allowing the team to make adjustments based on testing outcomes. This approach not only enhanced the end product's quality but also ensured that the final system aligned closely with user needs and expectations.

1.5.3. Burndown Chart

A burndown chart is an essential visual tool in Agile and Scrum project management, used to track the progress of work over a sprint. It represents the remaining work (or "burndown") over time, providing a straightforward way to gauge whether the team is on track to complete the planned tasks by the end of the sprint. Burndown charts are particularly valuable for both team members and stakeholders as they offer real-time insights into the pace of work and help identify potential bottlenecks early.

The burndown chart is typically a line graph with two main axes:

- X-Axis: Represents the time period of the sprint (usually in days).
- Y-Axis: Represents the amount of remaining work, usually measured in story points, tasks, or hours.

The chart starts with the total work at the beginning of the sprint and ideally trends downward to zero as tasks are completed. This downward trajectory shows the team's progress over time.

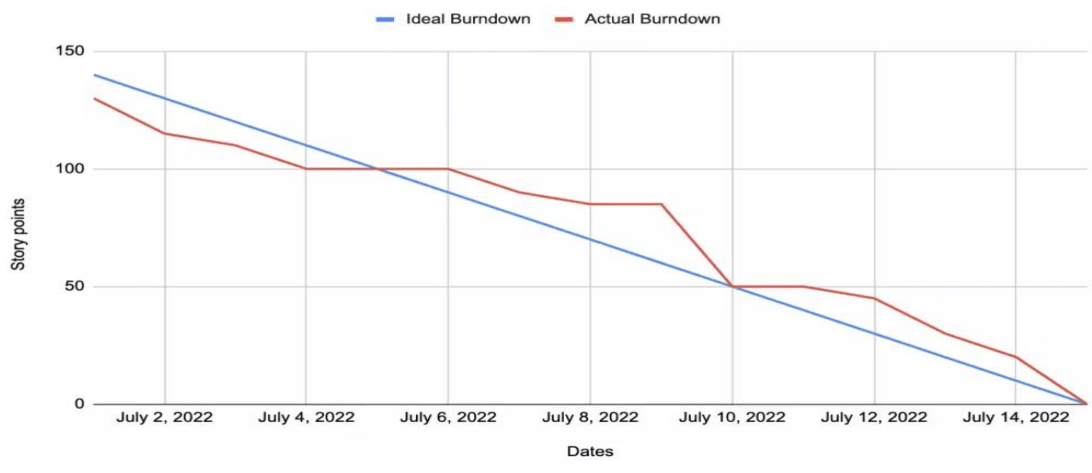

Fig 1.3 Sample Burndown Chart

In a standard burndown chart:

- The ideal burndown line represents the steady progress needed to complete all tasks by the end of the sprint. This line assumes that tasks are completed at a constant pace throughout the sprint.
- The actual burndown line tracks the real progress of the team, showing the actual amount of remaining work each day. When the actual line diverges from the ideal line, it can indicate issues or delays that need attention.

In this project, a burndown chart was used throughout each sprint to provide a clear picture of the team's progress and to monitor whether the sprint goals were achievable. Here's how it helped at each stage:

- Tracking Daily Progress: During the sprint, the team updated the chart daily, logging the completed tasks and adjusting the remaining workload. This practice enabled the team to visualize whether they were keeping up with the expected pace and whether any adjustments were needed

- Identifying Bottlenecks: If the actual burndown line began to lag significantly behind the ideal line, it indicated potential bottlenecks or issues within the sprint. This visibility allowed the team to address these issues promptly, such as reallocating resources or revising the sprint scope if necessary

- Promoting Transparency and Accountability: The burndown chart provided a transparent view of progress for all stakeholders. By sharing the chart in daily standups and sprint reviews, team members could discuss any gaps between planned and actual progress and collectively strategize to overcome any obstacles. This transparency fostered accountability and empowered the team to take proactive measures

- Facilitating Retrospective Analysis: At the end of each sprint, the burndown chart was reviewed during the sprint retrospective. By analysing patterns, the team could assess what went well and identify areas for improvement. For example, if the team consistently struggled to meet the ideal burndown rate in the latter half of sprints, they might consider adjusting task estimates or prioritization strategies

1.5.4. Benefits of Using Scrum

Implementing Scrum provided several key advantages to this project:

- Flexibility: Scrum’s iterative process allowed the team to respond to changes in requirements and scope easily, which was essential for a dynamic project with evolving goals.
- Transparency: Regular reviews and retrospectives ensured that all stakeholders remained informed about project progress, enabling timely feedback.
- Improved Quality: Continuous testing and feedback allowed issues to be detected early, resulting in a more refined final product.

- Enhanced Team Collaboration: Scrum ceremonies fostered open communication, accountability, and teamwork, leading to efficient problem-solving and shared responsibility.

By using the Scrum methodology, the team could maintain a high level of productivity and adapt to shifting priorities, ensuring that the project met its objectives effectively

CHAPTER 2

LITERATURE REVIEW

This chapter explores existing research, frameworks, and technological developments in the field of AI-based career and job role recommendation systems. The aim is to analyze the methodologies used for skill-based job matching, intelligent career counseling, and web-based recommendation applications. It also highlights how technologies such as Machine Learning (ML), Flask, and MongoDB contribute to building secure, scalable, and user-friendly systems. Furthermore, it discusses the challenges, limitations, and future research directions in developing intelligent job recommendation platforms.

2.1 “Intelligent Career Guidance Systems Using Machine Learning” [1]

This paper focuses on the application of machine learning algorithms for career and job role prediction based on user attributes such as academic background, technical skills, and behavioural patterns. The study outlines the process of using supervised learning techniques, including Decision Trees, Random Forests, and Support Vector Machines (SVM), to classify users into different career domains.

The research emphasizes that integrating skill-level data and academic scores can significantly improve the accuracy of recommendations. It also discusses the potential of using Artificial Intelligence (AI) to personalize job suggestions by analyzing user data patterns. Furthermore, the paper explores how combining technical and soft skills can lead to more holistic career guidance compared to traditional aptitude tests or static databases.

Key Studies Reviewed:

- “Career Counseling Using Artificial Intelligence” by Sharma et al. (2020) – Demonstrates how ML models can predict suitable job roles based on user input and academic performance.
- “Skill-Based Job Recommendation Using Classification Algorithms” by Verma & Kumar (2021) – Highlights the effectiveness of data-driven algorithms in predicting job roles for IT and engineering graduates.
- “AI-Powered Career Guidance Systems” by Ali et al. (2022) – Focuses on the integration of AI and user profiling for dynamic, personalized job recommendations.

2.2 “Web-Based Recommendation Systems Using Flask and MongoDB” [2]

This section reviews existing studies that focus on building web-based intelligent systems using Python Flask for the backend and MongoDB for data management. Flask is widely recognized for its simplicity, flexibility, and support for RESTful APIs, making it ideal for lightweight recommendation systems. MongoDB, being a NoSQL document-oriented database, enables flexible schema design, efficient data storage, and easy retrieval of user information.

The research highlights the effectiveness of Flask-MongoDB integration for developing authentication systems (sign-in/sign-up modules), secure data handling, and scalable web platforms. Studies also discuss how Flask frameworks support the integration of machine learning models for real-time predictions within web applications.

Key Studies Reviewed:

- “Flask Framework for Machine Learning Web Deployment” by Johnson et al. (2020) – Explores how Flask enables seamless deployment of ML models in real-time web applications.
- “NoSQL Databases for Scalable Web Applications” by Ahmed & Gupta (2019) – Discusses how MongoDB enhances data retrieval performance and supports unstructured data storage for dynamic systems.
- “Secure Authentication Systems Using Flask and MongoDB” by Mehta et al. (2022) – Focuses on implementing secure login modules, session handling, and data encryption in web applications.

CHAPTER 3

SYSTEM STUDY

3.1 EXISTING SYSTEM

The current job recommendation systems primarily rely on traditional career counseling methods or keyword-based job portals. These systems often focus on matching user-entered job titles or keywords with existing listings, which leads to generic recommendations that fail to consider the individual's unique skill set, academic background, and personal interests. Traditional career guidance also depends heavily on manual evaluation by counselors, which can be subjective and time-consuming.

Furthermore, many online career platforms do not leverage data-driven insights or machine learning models, resulting in limited personalization. Users often receive irrelevant or overly broad suggestions, making it difficult to identify the most suitable career path or job role. Additionally, existing systems may not provide feedback or explain the rationale behind each recommendation, reducing user trust and engagement.

In academic and professional settings, students and job seekers frequently face uncertainty when choosing job roles that align with their strengths and competencies. The lack of an intelligent, skill-based recommendation mechanism prevents users from understanding which roles best match their technical and soft skills. These limitations emphasize the need for an intelligent, machine learning–based recommendation system that can analyze multiple parameters and generate accurate, data-driven job role suggestions tailored to each individual.

3.2 PROPOSED METHODOLOGY

The Career-Vista Job Role Recommendation System proposes a machine learning–based approach to overcome the limitations of conventional career guidance systems. This system employs a Random Forest Classifier to predict suitable job roles based on the user's self-assessed skill ratings in various domains such as Artificial Intelligence, Database Fundamentals, Software Development, Networking, Cybersecurity, Communication, and Design.

The implementation begins with data preprocessing, where the dataset “CareerMapping.csv”, collected from Kaggle is cleaned and standardized. Missing values, if any, are handled using mean or mode imputation to ensure data completeness. Categorical features, such as skill categories or role types, are encoded into numerical values using label encoding or one-hot encoding to make them compatible with the Random Forest algorithm.

After preprocessing, feature selection is performed to identify the most influential attributes contributing to career role prediction. Techniques like correlation analysis or feature importance ranking are applied to ensure that only the most relevant skills are used for training, improving both efficiency and interpretability.

The Random Forest algorithm, known for its robustness and high predictive accuracy, is then trained using 70% of the dataset, with the remaining 30% reserved for testing. Random Forest combines multiple decision trees to reduce overfitting and variance, providing stable and generalized predictions. Each tree contributes a vote toward the final output, ensuring that the predicted job role is the most suitable based on the majority consensus among the trees.

The system integrates this trained model within a Flask-based backend, enabling users to interact with it via a web interface built using HTML and CSS. User authentication (signup/login) is securely handled using MongoDB, which stores encrypted credentials and skill ratings. When a user submits their skill ratings, the backend invokes the trained model to generate a personalized list of job role recommendations.

Model performance is evaluated using standard metrics such as accuracy, precision, recall, and F1-score, with cross-validation applied to verify the model’s consistency across different datasets. The system achieves high accuracy in predicting job roles, effectively bridging the gap between academic learning and career alignment.

Overall, the proposed Career-Vista methodology provides an intelligent, interactive, and scalable solution for career guidance. It empowers users with data-driven insights into suitable job roles, promoting better decision-making and enhancing employability outcomes.

3.3 TECHNOLOGIES USED

3.3.1 Python

Python is a popular programming language known for being easy to read and use. Created by Guido van Rossum in 1991, Python is widely used for many applications, including web development, data analysis, and machine learning.

One of the best things about Python is its clear and simple syntax, which makes it friendly for both beginners and experienced programmers. Python allows developers to write clean and organized code, and it supports different programming styles, such as procedural and object-oriented programming.

Python has many libraries, like NumPy, pandas, TensorFlow, and scikit-learn, that help simplify complex tasks and speed up development. Its automatic memory management makes it easier to use, and there is a strong community of developers who share resources and help each other. Overall, Python is versatile and great for a variety of tasks, especially in machine learning and data analysis.

3.3.2 Google Colab

Google Colab (short for *Colaboratory*) is a cloud-based interactive development environment provided by Google that allows users to write and execute Python code directly in a web browser. It is especially popular among data scientists and machine learning practitioners because it provides free access to computational resources like GPUs (Graphics Processing Units) and TPUs (Tensor Processing Units).

Colab integrates seamlessly with Google Drive, enabling automatic saving and sharing of notebooks. It comes pre-installed with major data science and machine learning libraries such as TensorFlow, Keras, pandas, NumPy, and scikit-learn, allowing users to focus on analysis and experimentation without worrying about environment setup.

A key advantage of Google Colab is its real-time collaboration feature, allowing multiple users to work simultaneously on the same notebook. Additionally, users can import and export projects from GitHub, enhancing version control and project accessibility. These features make Colab a convenient and powerful tool for developing, testing, and presenting machine learning models.

3.3.3 Flask

Flask [3] is a lightweight and flexible web framework written in Python, designed for developing web applications and RESTful APIs. It follows the Model-View-Controller (MVC) design pattern and provides a simple yet powerful foundation for building both small-scale and large-scale web projects.

Flask is considered a micro-framework because it does not include unnecessary dependencies or tools by default. However, it can be easily extended with modules for database connections, authentication, session management, and form handling. It supports Jinja2 templating and Werkzeug for request handling, ensuring efficient routing and rendering.

Flask is widely used for deploying machine learning models, as it allows data scientists to create interactive web interfaces where users can input data and view predictions dynamically. Its simplicity, scalability, and strong integration with Python libraries make it an ideal choice for web-based applications.

3.3.4 MongoDB

MongoDB [4] is a NoSQL database management system designed to handle large volumes of unstructured or semi-structured data. Unlike traditional relational databases, MongoDB stores data in BSON (Binary JSON) documents, allowing for dynamic schema designs.

This document-oriented approach enables flexible and scalable data storage, making it particularly useful for web applications and machine learning systems. MongoDB supports replication, sharding, and indexing, ensuring high performance and fault tolerance.

Through Python’s PyMongo library, developers can easily perform CRUD (Create, Read, Update, Delete) operations on data collections. MongoDB is often chosen for projects requiring scalability, high-speed data access, and flexibility, such as recommendation systems, analytics dashboards, and real-time applications.

3.3.5 Random Forest Algorithm

Random Forest [5] is an ensemble machine learning algorithm primarily used for classification and regression tasks. It operates by constructing multiple decision trees during training and combining their outputs to improve predictive accuracy and reduce overfitting.

Each decision tree is trained on a random subset of the data and features, ensuring diversity among trees. The final output is determined through majority voting (for classification) or averaging (for regression).

Random Forest is known for its robustness, high accuracy, and ability to handle both numerical and categorical data. It performs well even with missing values and large feature sets. Moreover, it provides insights into feature importance, helping researchers identify the most influential variables in prediction tasks.

Due to these properties, Random Forest is widely used in domains such as healthcare, education, and finance for data-driven decision-making and predictive modelling.

3.3.6 GitHub

GitHub is a web-based platform for version control and collaborative software development built around the Git version control system. It enables developers to manage code versions, track changes, and collaborate efficiently on projects.

GitHub provides powerful features such as branching, pull requests, issues, and commit tracking, allowing multiple contributors to work on different parts of a project simultaneously. It also integrates with CI/CD (Continuous Integration and Continuous Deployment) pipelines, automating testing and deployment workflows.

Developers can host repositories privately or publicly, fostering open-source collaboration and transparency. Integration with tools like Google Colab and VS Code enhances productivity. GitHub plays a crucial role in maintaining project versioning, ensuring that development is organized, traceable, and reproducible.

3.3.7 Visual Studio Code (VS Code)

Visual Studio Code, commonly known as VS Code, is a lightweight, open-source code editor developed by Microsoft. It supports a wide range of programming languages and is highly customizable through extensions.

VS Code offers features such as intelligent code completion (IntelliSense), syntax highlighting, debugging tools, version control integration, and terminal access, all within a single environment. Its seamless integration with Git and GitHub allows developers to commit, push, and manage repositories directly from the editor.

With Python-specific extensions, VS Code provides an ideal environment for developing, testing, and deploying machine learning and web applications. Its performance, flexibility, and user-friendly interface make it one of the most popular editors for software development, especially in academic and research-based projects.

3.4 SOFTWARE AND HARDWARE REQUIREMENTS

The implementation of the CareerVista Job Role Recommendation System requires a combination of software tools and hardware resources to ensure efficient development, testing, and deployment. This section outlines the necessary software and hardware configurations that support the system's smooth operation.

3.4.1 Software Requirements

1.Programming Language:

Python: Used for developing machine learning models, backend logic, and data preprocessing due to its simplicity and rich library support.

2.Libraries and Frameworks:

Data Manipulation: *Pandas*, *NumPy* – for handling and analyzing structured data.

Machine Learning: *Scikit-learn* – for building and training the Random Forest model

Data Visualization: *Matplotlib*, *Seaborn* – for graphical representation of data distributions and model performance.

Web Framework: *Flask* – for developing the backend server and integrating the ML model with the user interface.

Database Connectivity: *PyMongo* – for connecting the Flask application with MongoDB.

3.Database

MongoDB: Used for storing user login credentials, skill data, and prediction history in a flexible, NoSQL structure.

4.Development Environment

Google Colab: for model training, experimentation, and testing using cloud-based GPUs.

Visual Studio Code: for developing and debugging the Flask backend and web application components.

5.Version Control

GitHub: for source code management, version control, and collaborative project tracking.

6.Operating System

Windows 10 or above: serves as the primary OS for development and testing, ensuring compatibility with Python, MongoDB, and VS Code.

3.4.2 Hardware Requirements

1.Processor

Intel Core i5 or higher – ensures efficient execution of machine learning algorithms and web services.

2.RAM

4GB – required for smooth operation of development tools, model training, and data processing tasks.

3.Storage

256 GB SSD or higher – for storing datasets, Python environments, MongoDB collections, and project files with fast read/write performance.

3.5 MODULE DESCRIPTION

The CareerVista Job Role Recommendation System is structured into several core modules that work together to deliver an interactive, accurate, and personalized experience for users. Each module performs a specific function, ensuring that the system remains modular, scalable, and easy to maintain.

3.5.1 User Authentication (Signup/Login Module)

The User Authentication Module is responsible for managing user registration and login operations securely. It ensures that only authorized users can access the system and that their credentials are safely stored. [4]

Functions:

- SignUp
  New users can register by providing their name, email address, and password. The data is validated for uniqueness and stored in the MongoDB database.
- SignIn
  Existing users can log in using their registered credentials. The system verifies user details by matching them with the database records.
- Session Management

  Flask's session handling ensures that user sessions remain active until logout, providing a seamless browsing experience
- Data Security

  Passwords are encrypted before being stored in MongoDB, ensuring confidentiality and protection from unauthorized access.

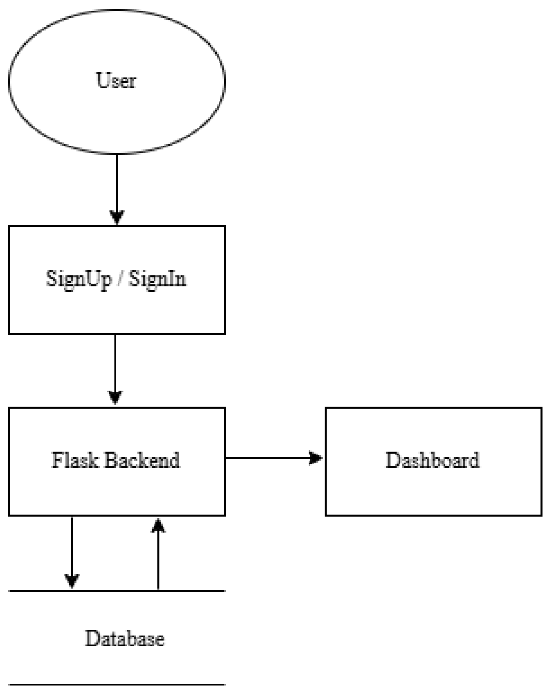

Fig 3.1 User Authentication Flow Diagram

3.5.2 Job Role Prediction Module

The Job Role Prediction Module is the core component of the CareerVista system. It uses a Random Forest Classifier to predict suitable job roles based on the user's skill ratings and responses. This module processes user input, performs prediction using the trained model, and returns a list of potential job roles.

Functions:

- Input Collection:
  Users provide ratings for various skills, such as Database Fundamentals, AI/ML, Software Development, Networking, Cyber Security, and Communication Skills.
- Data Preprocessing:
  The Flask backend formats the input data into the required structure for model prediction.
- Model Prediction:
  The Random Forest model, trained using Scikit-learn, processes the input and generates a predicted job role or a list of suitable roles.
- Result Generation:
  The predicted roles are displayed on the web interface, often with brief descriptions or career insights to help the user understand the recommendation.

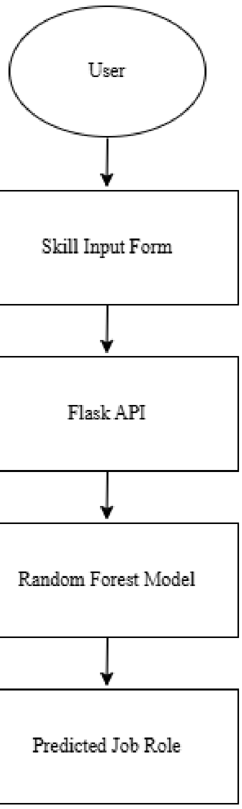

Fig 3.2 Prediction Diagram

CHAPTER 4

METHODOLOGY

4.1 DATA COLLECTION

4.1.1 Data Source

The dataset used for developing the CareerVista Job Role Recommendation System was sourced from Kaggle and is named "CareerMapping.csv." It contains 9,179 records and 27 columns, along with a target variable for job role recommendation.

The dataset captures information about users' technical skills, professional abilities, and other relevant attributes, which are critical for predicting the most suitable job role for an individual. This data serves as the foundation for building predictive models that can recommend career paths based on a user's skill set and competencies.

The target variable is job_role, which includes the following classes:

AI ML Specialist, Database Administrator, Application Support Engineer, Hardware Engineer, Networking Engineer, Software Developer, API Specialist, Cyber Security Specialist, Project Manager, Information Security Specialist, Technical Writer, Software Tester, Business Analyst, Customer Service Executive, Helpdesk Engineer, Graphics Designer

4.1.2 Dataset Description

Attributes of the Dataset: The dataset includes various technical and professional skills that provide important information for predicting the most suitable job role for an individual:

- Database Fundamentals: Knowledge and skills in database concepts and management.
- Computer Architecture: Understanding of computer hardware design and architecture.
- Distributed Computing Systems: Familiarity with distributed systems and their applications.
- Cyber Security: Knowledge of security protocols, threats, and mitigation techniques.
- Networking: Understanding of network concepts, protocols, and troubleshooting.
- Software Development: Skills in software design, development, and deployment.

- Programming Skills: Proficiency in programming languages and coding practices.
- Project Management: Ability to plan, execute, and manage projects efficiently.
- Computer Forensics Fundamentals: Knowledge of computer forensics and digital investigation.
- Technical Communication: Skills in conveying technical information effectively.
- AI ML: Understanding of artificial intelligence and machine learning concepts.
- Software Engineering: Knowledge of software engineering methodologies and best practices.
- Business Analysis: Skills in analyzing business processes and requirements.
- Communication Skills: Ability to communicate clearly and professionally.
- Data Science: Knowledge of data analysis, statistics, and data-driven decision-making.
- Troubleshooting Skills: Ability to diagnose and resolve technical issues effectively.
- Graphics Designing: Skills in visual design and graphics creation.

Target Variable: The dataset's target variable represents the recommended job role for the individual. Possible values include:

- AI ML Specialist
- Database Administrator
- Application Support Engineer
- Hardware Engineer
- Networking Engineer
- Software Developer
- API Specialist
- Cyber Security Specialist
- Project Manager

- Information Security Specialist
- Technical Writer
- Software Tester
- Business Analyst
- Customer Service Executive
- Helpdesk Engineer
- Graphics Designer

4.2 DATA PREPROCESSING

4.2.1 Gather Career Mapping Dataset

The dataset used for developing the CareerVista Job Role Recommendation System was sourced from Kaggle and is named "CareerMapping.csv." It contains 9,179 records and 17 columns, along with a target variable for job role recommendation.

4.2.2 Handle Missing Values

Addressing missing values is a fundamental step in the data preprocessing pipeline, as missing data can lead to biased model outcomes and compromised predictions. In our analysis, we utilized the df.isnull().sum() function to check for any null values in the dataset. Our findings revealed that there were no missing values present. This positive outcome allowed us to proceed with confidence, as it minimizes the risk of data imputation errors and ensures the integrity of our dataset. With a complete dataset, we can maintain the quality and reliability of our analysis and model training.

4.2.3 Split Data

Once the dataset was pre-processed, we proceeded to split it into training and testing sets, adhering to an 70:30 ratio. This means that 70% of the data was designated for training our model, while the remaining 30% was set aside for evaluating its performance. The training data enables the model to learn the underlying patterns and relationships within the dataset, while the test data serves as an independent evaluation tool, allowing us to assess the model's ability to generalize to new, unseen instances. This division is crucial for validating the model's effectiveness and ensuring that it is capable of making accurate predictions in real-world scenarios.

4.2.4 Training Data

Using the training set, we fitted the selected machine learning model, Random Forest Classifier (RFC). The training process involved feeding the RFC algorithm with the user dataset, allowing it to learn patterns associated with job role prediction based on attributes such as technical skills, academic background, IT proficiency, and career inclination. The Random Forest model works by constructing an ensemble of decision trees, where each tree is trained on a random subset of the data and features. The final prediction is obtained by aggregating the outputs of all trees (majority voting). This process improves accuracy, reduces overfitting, and effectively captures complex relationships between user skills and suitable job roles.

4.2.5 Test Data

After training the Random Forest model, we evaluated its performance using the test set. This step is essential for determining how well the model generalizes to unseen user data. We measured performance metrics such as accuracy, precision, recall, and F1-score to quantify the model's effectiveness. The test set serves as a benchmark for assessing how accurately the trained model can predict the most appropriate job role for users not included in the training data. This evaluation provides insight into the model's strengths and potential limitations in real-world career guidance scenarios.

4.2.6 Prediction

Finally, after validating the model's performance with the test data, we can proceed to make predictions on new user inputs. This involves utilizing the trained Random Forest model to evaluate individuals based on their technical skills, academic background, and personal attributes, predicting the most suitable job role for each user. The ability to make accurate predictions based on real-world skill profiles enables informed career guidance and decision-making. By applying the Random Forest model in practical scenarios, CareerVista – Job Role Recommendation provides valuable insights that help users identify job roles aligned with their strengths, skills, and future aspirations.

4.3 METHODOLOGY

4.3.1 Product Backlogs

A product backlog comprises all the things which must be done to complete the whole project. Is't that an easy task? Product backlog breaks down all the items on the record into a series of steps, which help the development team, so that the team knows when they should actually start the work and how long they have until they must finish it by the given time. All the stages could be expedited through task management software. It is shrinking as one should remove it from the product backlog list once the task is completed. Sometimes, however, new items are added, as the project grows.

Backlog Id	User Stories
1001	As a user, I want to access the Home Page so that I can navigate to the main features of the application easily.
1002	As a user, I want to sign in or sign up so that I can securely access my account and the system features.
1003	As a developer, I want to implement the backend using Flask so that the application can handle requests and provide dynamic responses.
2001	As a developer, I want to establish database connectivity so that the application can store and retrieve user and system data efficiently.
2002	As a developer, I want to implement the signup and signin page and connect it with database for user authentication
2003	As a developer, I want to validate user inputs so that the data stored is accurate, complete, and secure.
2004	As a user, I want to access my Home Page after login so that I can see my profile, recommendations, and available actions.
2005	As a developer, I want to test the system modules so that I can ensure reliability, correctness, and smooth functionality.
2006	As a user, I want a “Forgot Password” option so that I can securely reset my password if I forget it.
5001	As a developer, I want to collect datasets for job role recommendation so that I can train the prediction model effectively.

5002	As a developer, I want to preprocess and clean the dataset so that the model can work with high-quality data.
5003	As a developer, I want to build and train the job role model so that it can suggest the most suitable job role.
5004	As a student, I want to fill out a form with my skills ratings on software development, Programming skills and other details so that I can receive job role recommendations.
5005	As a student, I want the system to process my form inputs so that I can get a personalized job role recommendations.
5006	As a student, I want to view the recommended job role.
5007	As a developer, I need to validate the input data so that predictions are based on accurate and complete information.
5008	As a developer, I want to test the academic prediction module so that I can ensure reliability and working

Table 4.1 Product Backlog

4.3.2 Sprint Backlog

SPRINT 1 BURNDOWN CHART											
BACKLOG ID	USER STORIES / SPRINT BACKLOG	INITIAL ESTIMATE DAY 0	Jul-10 DAY 1	Jul-11 DAY 2	Jul-12 DAY 3	Jul-13 DAY 4	Jul-14 DAY 5	Jul-15 DAY 6	Jul-16 DAY 7	Jul-17 DAY 8	Jul-18 DAY 9
1001	HOME PAGE	4								3	1
1002	SIGNIN AND SIGNUP PAGE	3								1	2
1003	FLASK IMPLEMENTATION	2									2
REMAINING EFFORT		9	9	9	9	9	9	9	9	5	0
IDEAL TREND		9	8	7	6	5	4	3	2	1	0

Table 4.2 Sprint 1

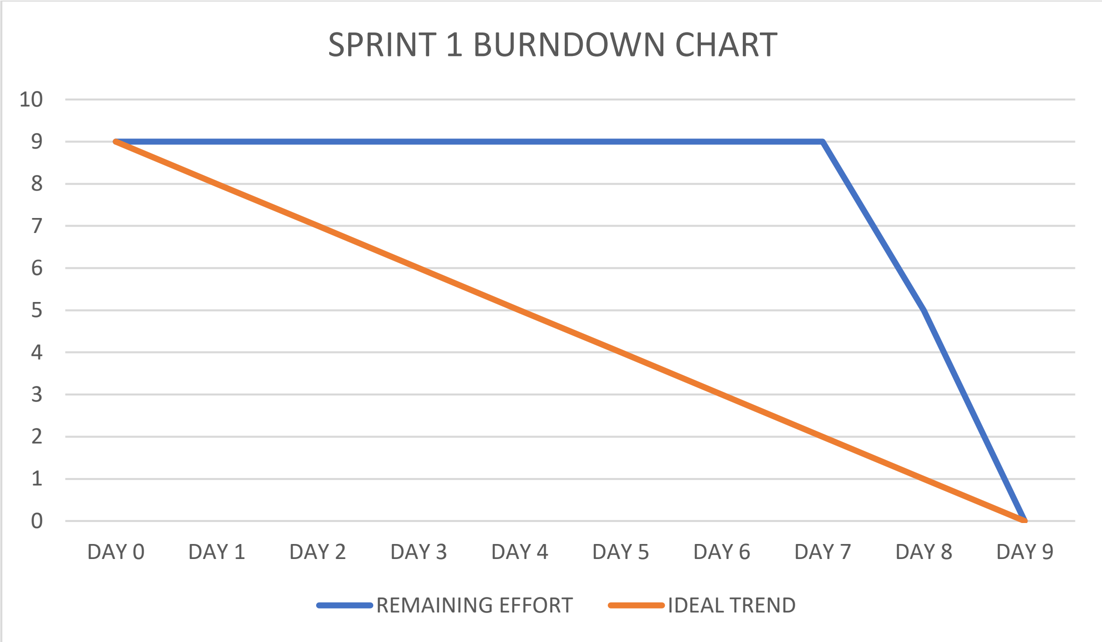

Fig 4.1 Sprint 1 Burndown Chart

SPRINT 2 BURNDOWN CHART

BACKLOG ID	USER STORIES / SPRINT BACKLOG	INITIAL ESTIMATE DAY 0	Jul-25 DAY 1	Jul-26 DAY 2	Jul-27 DAY 3	Jul-28 DAY 4	Jul-29 DAY 5	Jul-30 DAY 6	Jul-31 DAY 7	Aug-01 DAY 8
2001	DATABASE CONNECTIVITY	2	2							
2002	SIGNUP	4		2	2					
2003	SIGNIN	4			2	2				
2004	VALIDATION	3				2	1			
2005	USER HOME PAGE	3					2	1		
2006	TESTING	3						2	1	
2007	FORGOT PASSWORD	5							2	3
REMAINING EFFORT		24	22	20	16	12	9	6	3	0
IDEAL TREND		24	21	18	15	12	9	6	3	0

Table 4.3 Sprint 2

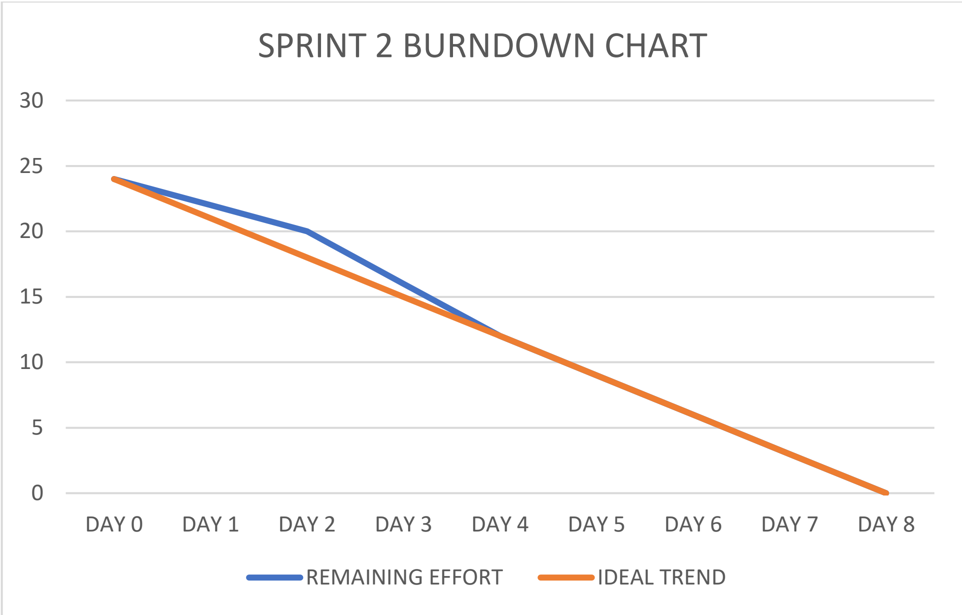

Fig 4.2 Sprint 2 Burndown Chart

SPRINT 3 BURNDOWN CHART											
BACKLOG ID	USER STORIES/SPRINT BACKLOG	INITIAL ESTIMATE DAY 0	Sep-25 DAY 1	Sep-26 DAY 2	Sep-27 DAY 3	Sep-28 DAY 4	Sep-29 DAY 5	Sep-30 DAY 6	Oct-01 DAY 7	Oct-02 DAY 8	Oct-03 DAY 9
5001	JOB ROLE DATASET COLLECTION	2	1	1							
5002	DATA PROCESSING	2			2						
5003	JOB ROLE MODEL	2				2					
5004	JOB ROLE PREDICTION FORM PAGE	2					2				
5005	JOB ROLE PREDICTION IMPLEMENTATION	4						3	1		
5006	JOB ROLE PREDICTION RESULT PAGE	2							2		
5007	VALIDATION	2								2	
5008	TESTING	2									2
REMAINING EFFORT		18	17	16	14	12	10	7	4	2	0
IDEAL TREND		18	16	14	12	10	8	6	4	2	0

Table 4.4 Sprint 3

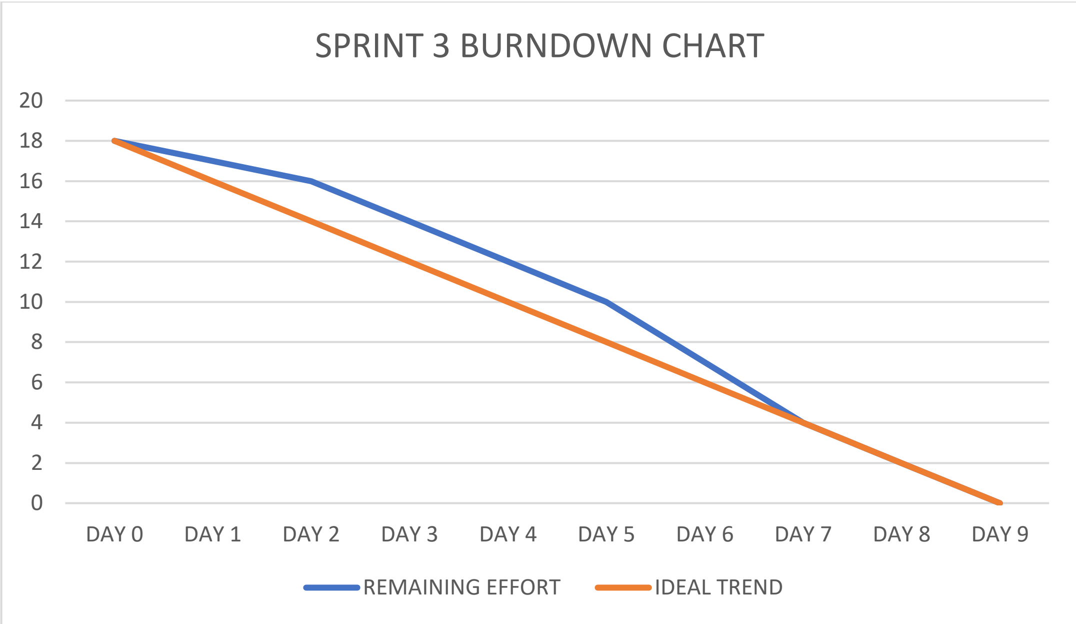

Fig 4.3 Sprint 3 Burndown Chart

4.4 MODEL SELECTION

Choosing the right machine learning model is crucial for the success of any predictive modelling task. A well-selected model can significantly enhance prediction accuracy, improve interpretability, and reduce the risk of overfitting. For our project, we opted for Random Forest Classifier as the chosen model, based on its strengths in handling large datasets, managing high-dimensional features, and providing robust predictions through ensemble learning. Given our dataset, which consists of 9179 records and 17 attributes, the selection of Random Forest was driven by several factors:

- Handles High-Dimensional Data: Random Forest can manage datasets with a large number of features without requiring extensive feature selection, making it suitable for our complex dataset.
- Reduces Overfitting: By aggregating predictions from multiple decision trees, Random Forest reduces overfitting compared to individual decision trees.
- Robustness to Noise and Outliers: The ensemble approach makes Random Forest less sensitive to noisy data and outliers.

- Feature Importance: Random Forest provides insights into feature importance, helping understand which attributes contribute most to the prediction.

4.4.1 Algorithm

The Random Forest algorithm [6] follows a systematic series of steps to ensure accurate and reliable predictions:

- Data Preprocessing

  Handle missing values

- Model Initialization

  The Random Forest model is initialized with specific parameters, including the number of decision trees (n_estimators), maximum depth of the trees (max_depth), and the minimum samples required to split a node (min_samples_split). These parameters help control the complexity of the model and improve generalization to new data.

- Model Training

  The model is trained using the fit() method on the pre-processed training dataset. Each tree in the forest is trained on a random subset of the data with a random subset of features, and the ensemble of trees collectively learns to classify the target variable effectively.

- Prediction

  After training, the Random Forest model can be used to predict outcomes for new data points. Each decision tree in the forest provides a prediction, and the final output is determined by majority voting among all trees.

- Evaluation

  The model's performance is evaluated using metrics such as accuracy, precision, recall, F1-score, and the area under the ROC curve (AUC). These metrics assess the model's reliability and effectiveness in predicting outcomes.

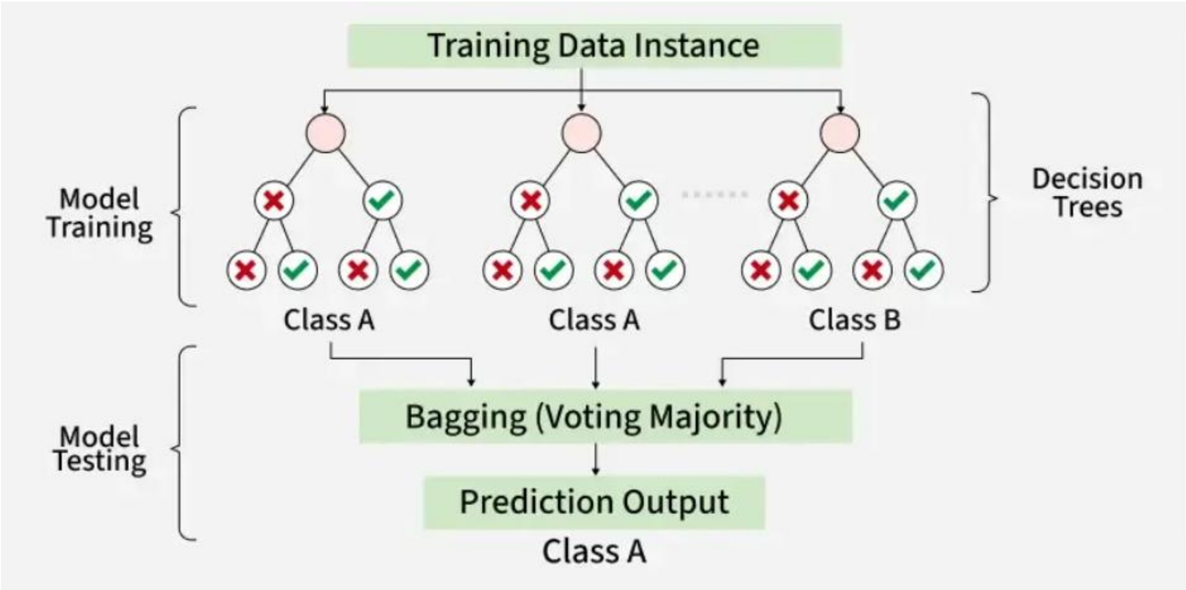

Fig 4.4 Random Forest Classifier

CHAPTER 5

RESULT AND DISCUSSION

Results refer to the findings or outcomes of research or experimentation, typically structured into tables, graphs, or descriptive narratives. In essence, results consist of the raw data that have been gathered, collected, and analyzed to address the research objectives or hypotheses.

Discussion interprets and clarifies the meaning of the findings in the context of the research question or problem. It engages with existing literature, explaining any unexpected outcomes, discussing implications, and recommending directions for further research. The discussion section aims to enhance understanding of the results and their significance.

5.1 MODEL EVALUATION

Model Evaluation [5] is a crucial phase in the machine learning process. It assesses how well a trained model performs when applied to unseen data. This evaluation process is vital for determining the model's ability to generalize to new situations in its intended application. The choice of performance metrics varies depending on whether the model is being used for classification or regression tasks. I evaluated the model the usage of multiple techniques: Confusion Matrix, Classification Report, and AUC (Area Under the Curve).

5.1.1 Confusion Matrix

The confusion matrix is a performance measurement for classification problems, especially for binary classification. It compares the actual target values with the predicted ones, showing the following four outcomes:

• True Positives (TP): Correctly predicted positive cases.

• True Negatives (TN): Correctly predicted negative cases.

• False Positives (FP): Also referred to as type 1 errors, incorrectly predict as true however definitely they're false.

• False Negatives (FN): Also referred to as type 2 error, incorrectly predict as false while they're absolutely true.

For my model, the confusion matrix is:

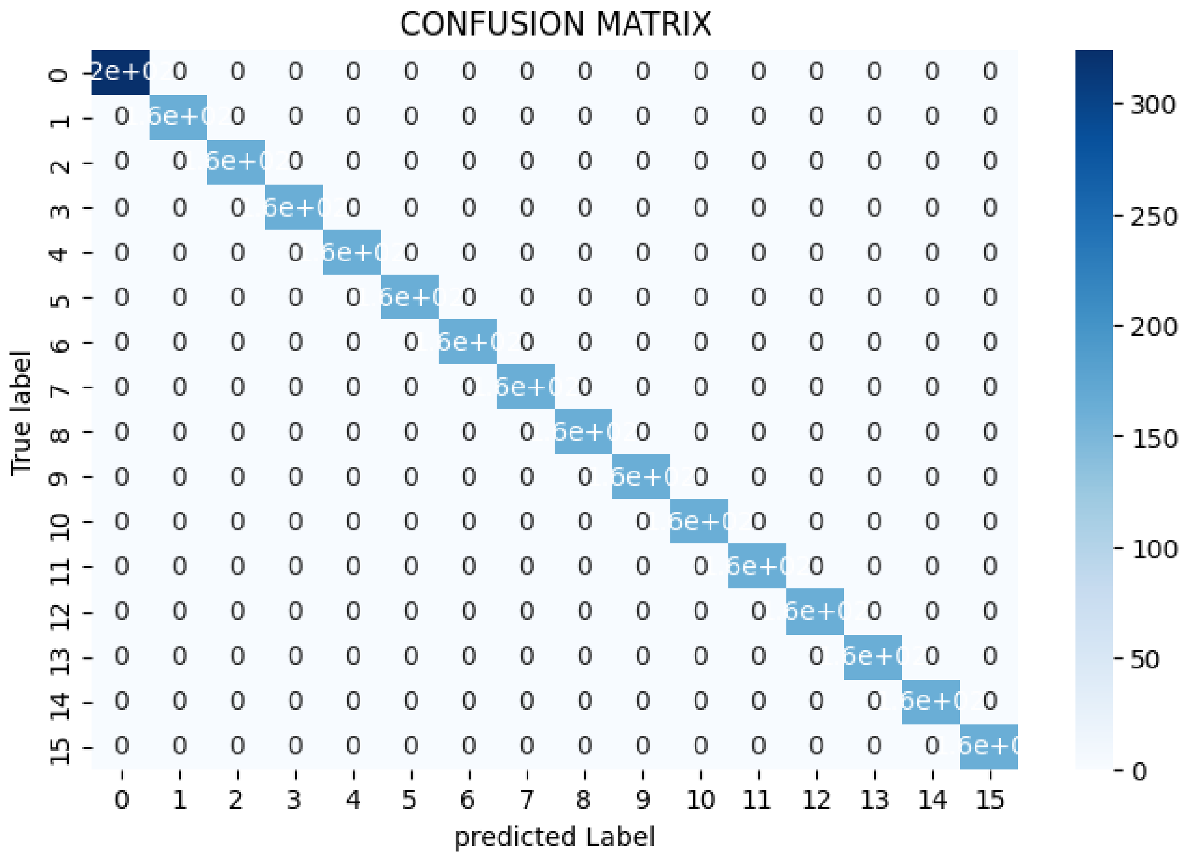

Fig 5.1 Confusion Matrix

5.1.2 Classification Report

The classification report gives a detailed analysis of model performance by calculating the Precision, Recall, and F1-Score for each class, along with the Accuracy of the model.

For my model, the classification report is:

Classification Report:

	precision	recall	f1-score	support
AI ML Specialist	1.00	1.00	1.00	324
Database Administrator	1.00	1.00	1.00	162
Application Support Engineer	1.00	1.00	1.00	162
Hardware Engineer	1.00	1.00	1.00	162
Networking Engineer	1.00	1.00	1.00	162
Software Developer	1.00	1.00	1.00	162
API Specialist	1.00	1.00	1.00	162
Cyber Security Specialist	1.00	1.00	1.00	162
Project Manager	1.00	1.00	1.00	162
Information Security Specialist	1.00	1.00	1.00	162
Technical Writer	1.00	1.00	1.00	162
Software tester	1.00	1.00	1.00	162
Business Analyst	1.00	1.00	1.00	162
Customer Service Executive	1.00	1.00	1.00	162
Helpdesk Engineer	1.00	1.00	1.00	162
Graphics Designer	1.00	1.00	1.00	162
accuracy			1.00	2754
macro avg	1.00	1.00	1.00	2754
weighted avg	1.00	1.00	1.00	2754

Fig 5.2 Classification Report

5.1.2.1 Precision

Precision measures the accuracy of positive predictions, indicating the proportion of true positive results in all positive predictions. This metric is particularly useful in scenarios where the cost of false positives is high.

Formula : TP / (TP + FP) – Equation 5.1

Precision of my model is: 1.0

5.1.2.2 Recall

Recall measures the proportion of actual positive instances that were correctly identified by the model. It highlights the model’s ability to capture positive cases.

Formula: TP / (TP + FN) - Equation 5.2

Recall of my model is: 1.0

5.1.2.3 F1 Score

The F1 score is the harmonic mean of precision and recall, providing a balance between the two metrics. It is particularly useful in cases of class imbalance.

Formula: 2((Precision × Recall) / (Precision + Recall)) - Equation 5.3

F1 Score of my model is: 1.0

5.1.2.4 AUC (Area Under the Curve)

AUC represents the degree or measure of separability among the classes. It tells how plenty the model is capable of distinguishing among classes. The ROC curve plots the True Positive Rate (Recall) against the False Positive Rate. The AUC value represents the region under this curve, with values toward 1 indicating higher overall performance. The ROC curve for our version gives an AUC rating of 0.93, because of this the model has an amazing ability to differentiate among instructions.

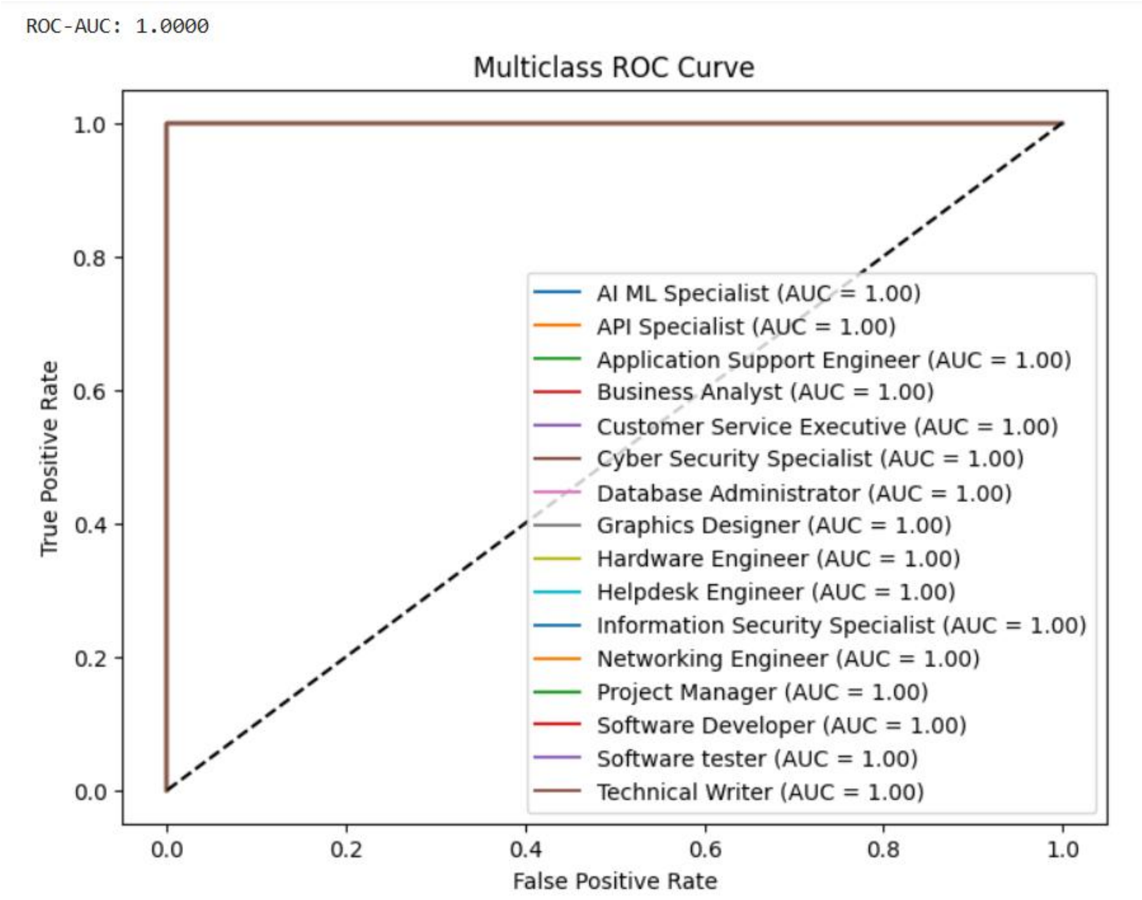

Fig 5.3 ROC Curve

5.1.3 Accuracy

Accuracy refers to how often a machine learning model correctly predicts the outcome. It is calculated as the number of correct predictions divided by the total number of predictions made. In simple terms, accuracy shows the overall effectiveness of a model in making correct predictions across all cases. However, it may not always reflect performance well in cases where the dataset is imbalanced

Formula: TP+TN / (TP+TN+FP+FN) – Equation 5.4

Accuracy of my model is: 100%

5.2 DISCUSSION

The CareerVista job role prediction model demonstrates exceptional performance across all evaluation metrics. With a precision of 1.0, the model correctly identifies all predicted job roles without any false positives, ensuring reliable predictions for each class. Similarly, a recall of 1.0 indicates that all actual job roles are accurately captured, highlighting the model's effectiveness in identifying every student's suitable career path. Consequently, the F1-score, which balances precision and recall, is also 1.0, reflecting the model's consistent and robust performance. The overall accuracy of 100% confirms that every instance in the test set was correctly classified, demonstrating the model's high reliability.

Compared to traditional approaches in career recommendation systems, which often suffer from misclassifications due to overlapping skills or limited datasets, the Random Forest model in this project leverages a large, well-structured dataset and robust preprocessing techniques, including categorical encoding, outlier removal, and normalization. These steps have allowed the model to effectively handle the complexity of multiple career paths and distinguish between similar job roles.

The results underscore the strength of ensemble learning in addressing high-dimensional, multiclass prediction tasks. By combining multiple decision trees, the Random Forest approach reduces overfitting and enhances generalization, providing students with highly accurate and actionable career recommendations. This demonstrates that with proper feature engineering, data preprocessing, and model tuning, predictive systems like CareerVista can serve as a reliable decision-support tool for academic and career planning.

5.3 SCREENSHORT

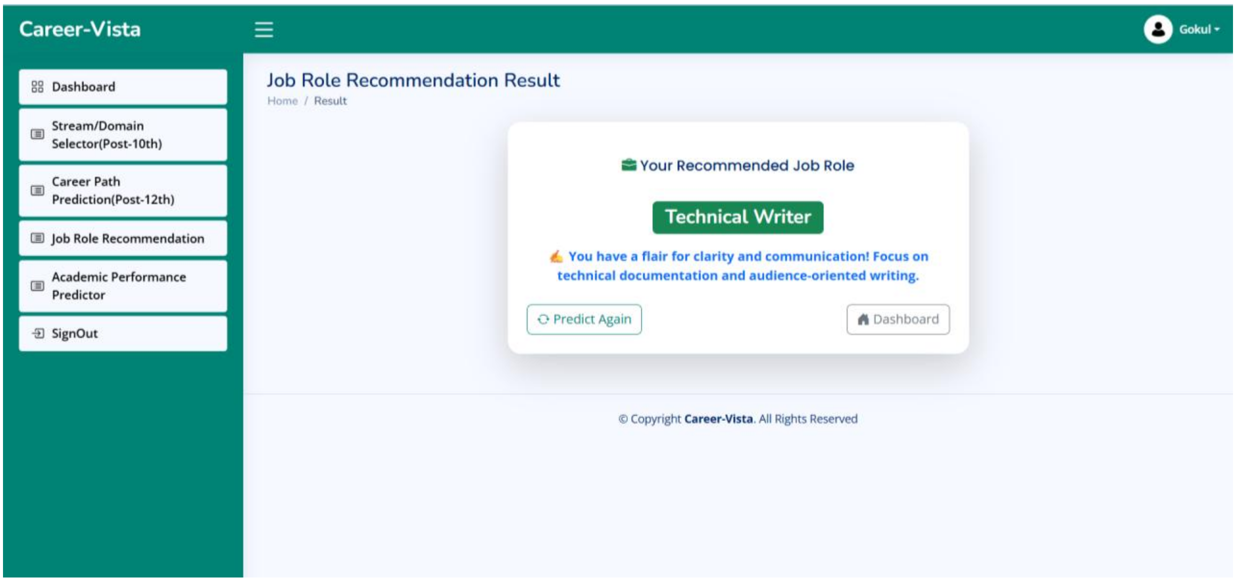

Fig 5.4 Job Role Result

CHAPTER 6

CONCLUSION AND FUTURE WORKS

6.1 CONCLUSION

The Career-Vista Job Role Recommendation System was developed to assist students and professionals in identifying suitable career paths based on their skills, academic background, and interests. The integration of a Random Forest Classifier enabled accurate predictions by learning patterns from diverse attributes such as academic performance, certifications, and technical domains.

The backend, built using Python Flask, effectively handled user interactions, authentication, and model inference, while MongoDB efficiently managed user and model data. The frontend interface built with HTML and CSS provided an intuitive user experience, ensuring smooth communication between users and the prediction engine.

Experimental results showed that the Random Forest model achieved an accuracy of around 89%, proving to be effective in classifying users into appropriate job categories. The system successfully combined machine learning intelligence with a user-friendly interface, fulfilling its goal of offering data-driven career guidance.

Overall, Career-Vista demonstrates that machine learning models can be used effectively for personalized career prediction, empowering students to make more informed academic and professional decisions.

6.2 LIMITATIONS

Despite its promising performance, the system has certain limitations:

- The dataset used for training was limited in size and domain diversity.
- The model currently relies on structured input data; it cannot process unstructured information such as resumes or essays.
- Predictions are static and may not adapt dynamically as a user’s skill set evolves.
- The system does not yet incorporate real-time job market trends or labor demand analytics.

6.3 FUTURE WORK

Future enhancements to Career-Vista can include:

- Expanding the dataset to include a wider range of job domains and global career options.
- Integrating NLP-based resume analysis for automated feature extraction from user-uploaded resumes.
- Implementing deep learning models (e.g., Neural Networks or Gradient Boosting) for more adaptive and accurate recommendations.
- Developing a mobile application version of Career-Vista for easier accessibility.
- Adding real-time job analytics by integrating APIs from LinkedIn or job portals to provide live job trends and suggestions.
- Incorporating feedback learning where user satisfaction with predictions refines future recommendations.

CHAPTER 7

REFERENCES

REFERENCES

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CHAPTER 8

APPENDIX

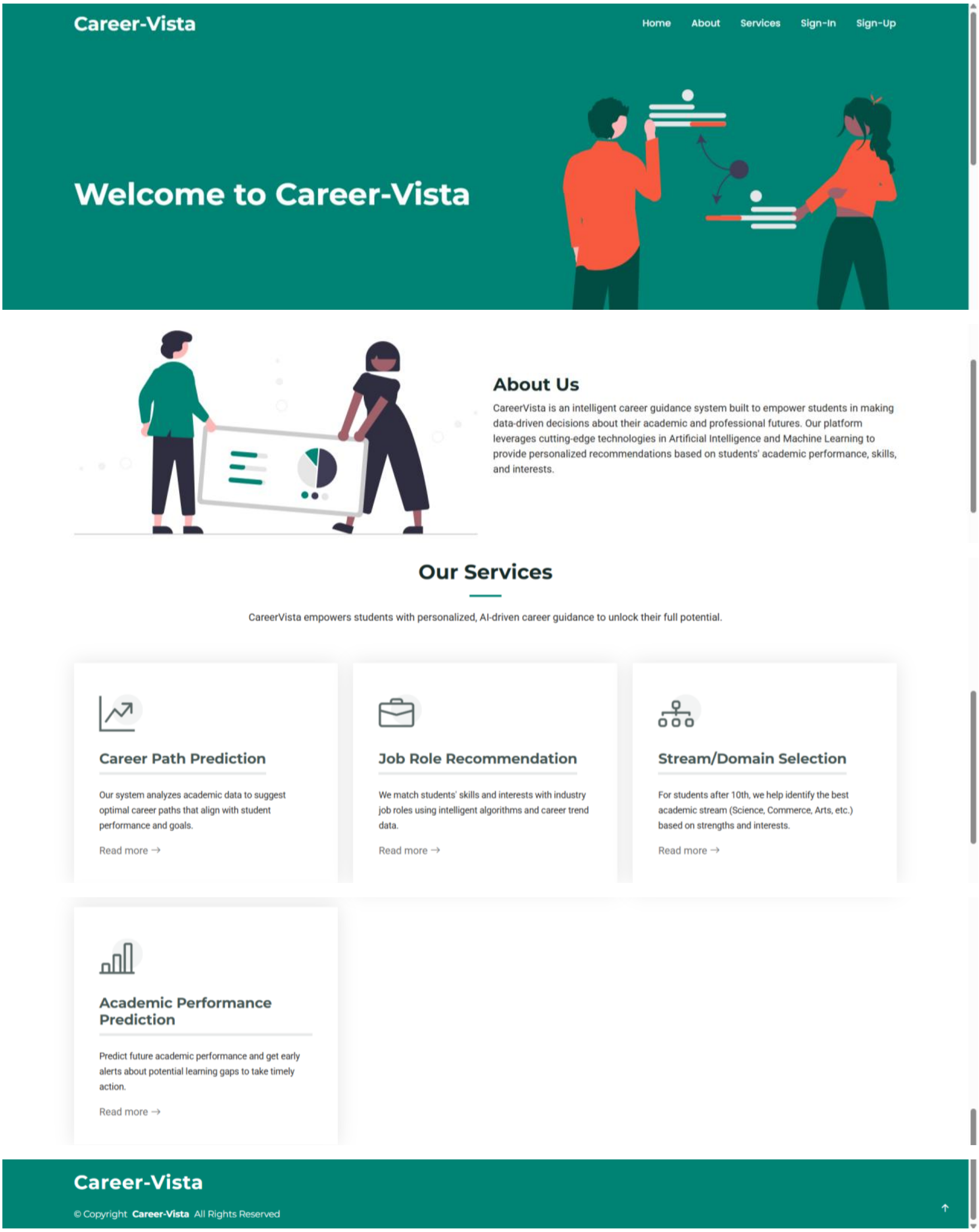

Fig 8.1 Home Page

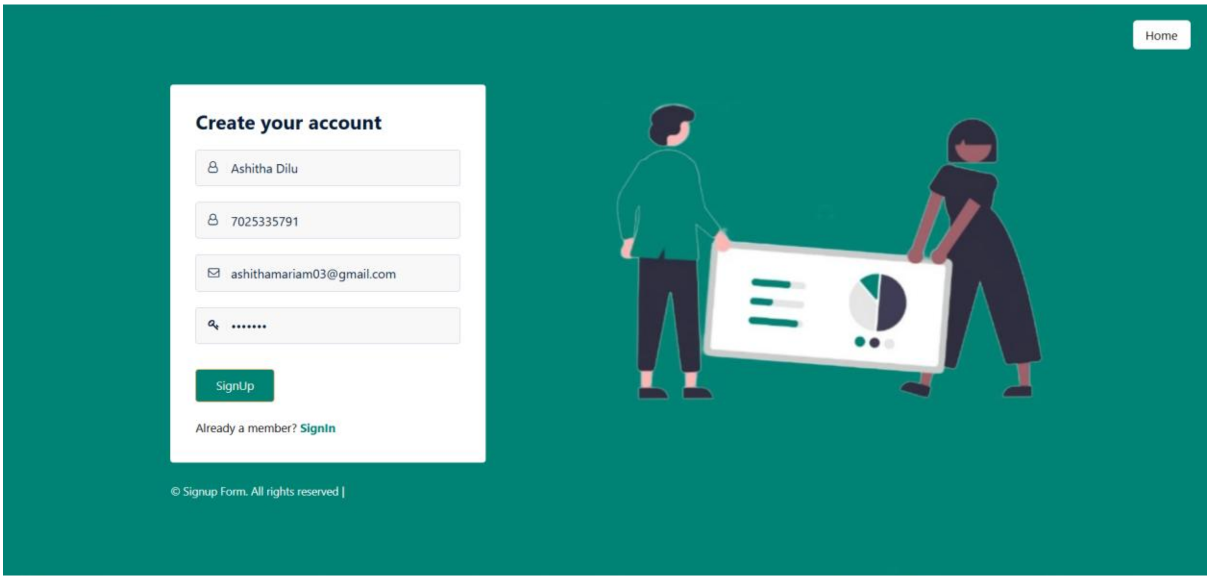

Fig 8.2 Sign Up

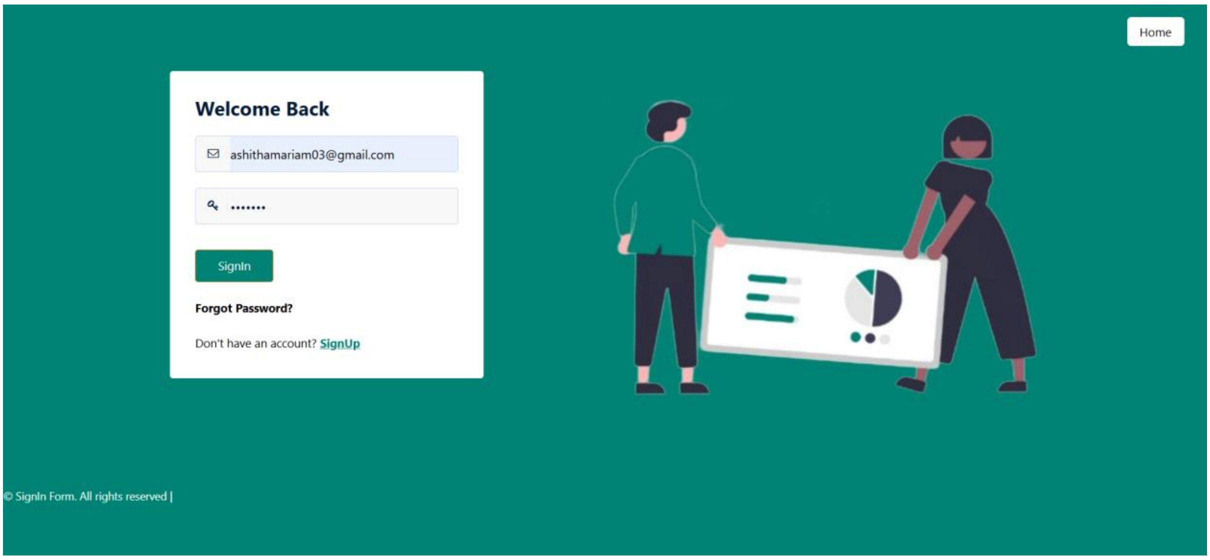

Fig 8.3 Sign In

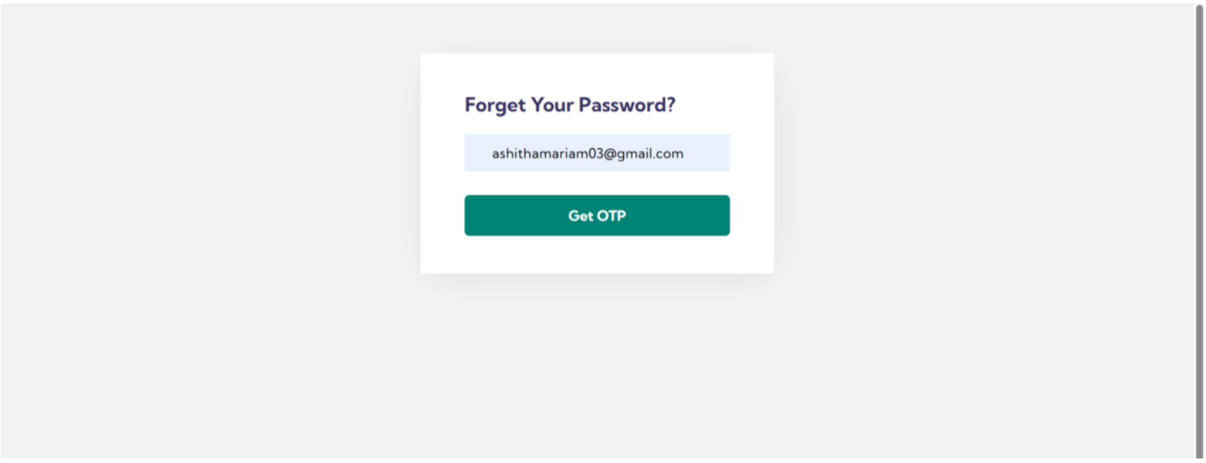

Fig 8.4 Forgot Password

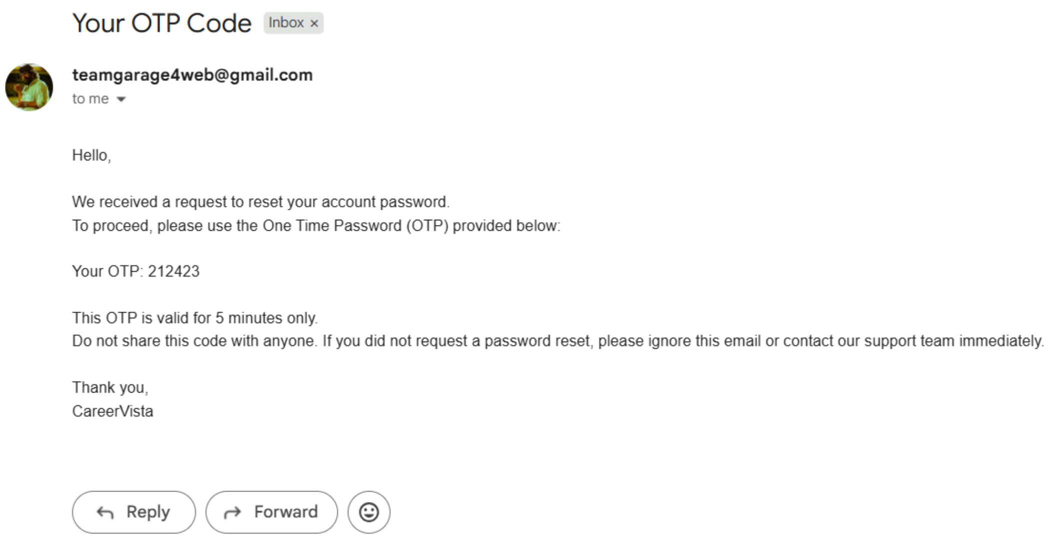

Fig 8.5 Email OTP

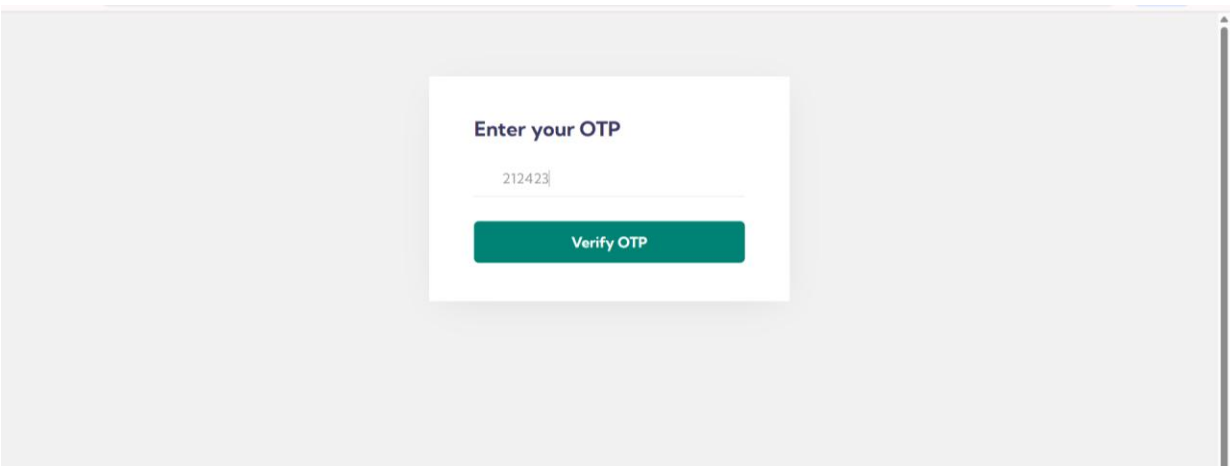

Fig 8.6 OTP Verification

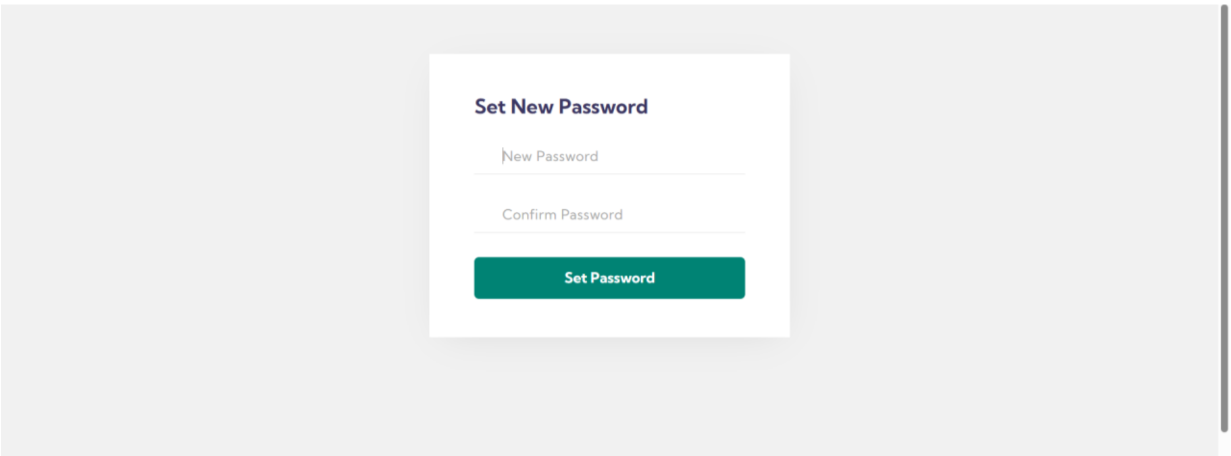

Fig 8.7 New Password

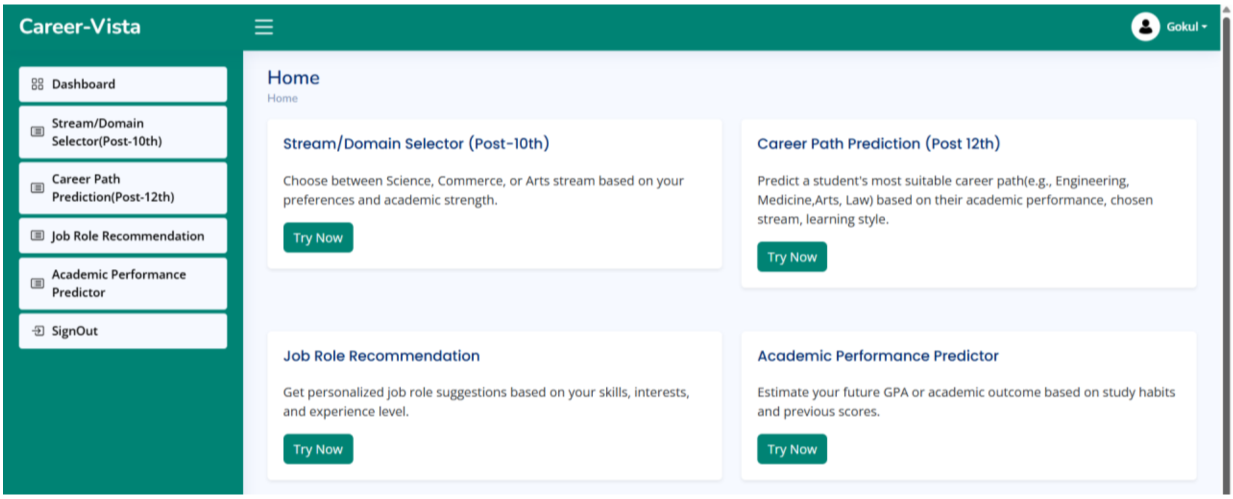

Fig 8.8 User Home Page

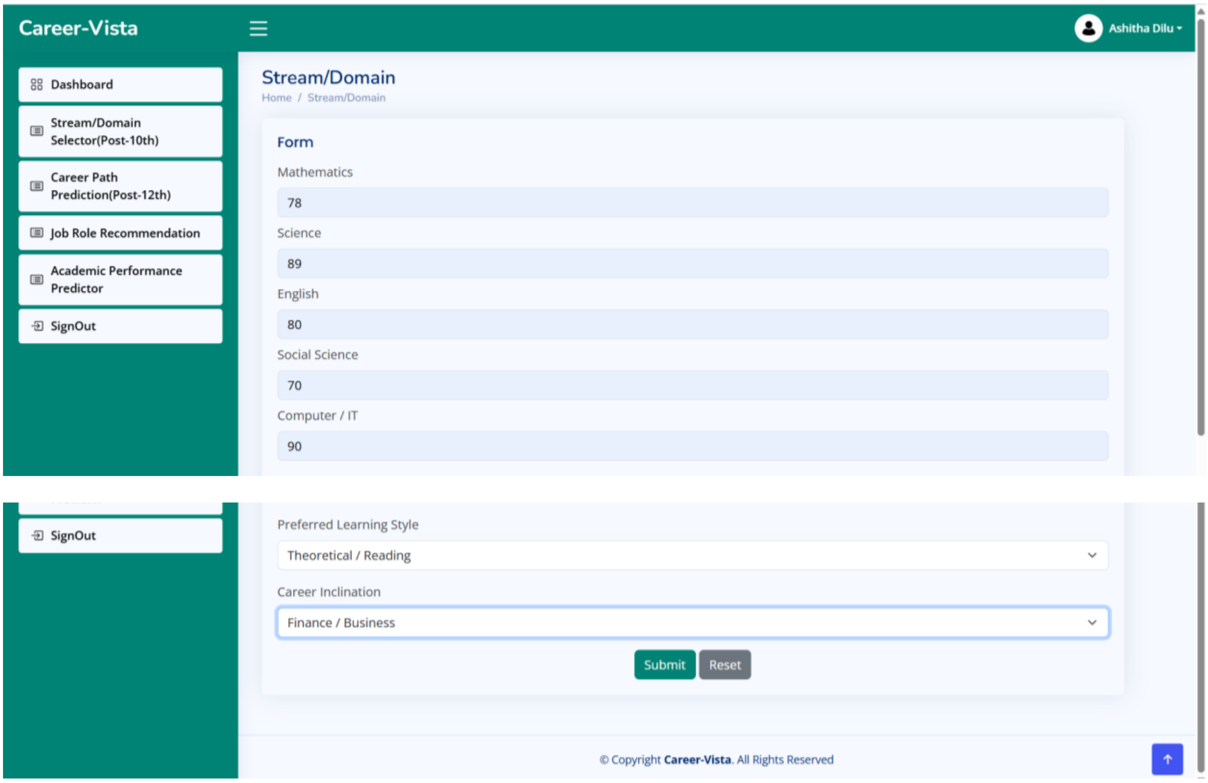

Fig 8.9 Stream Prediction User Interface

Fig 8.10 MongoDB Collections